DSC5002 Project Proposal

Course: Exploratory Data Analysis and Visualization

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Title of the Project

Missing Value Handling and Visual Interpretation of Chronic Kidney Disease Data

Introduction

Clinical datasets contain numerical measurements, categories and missing values. Removing incomplete records or filling missing entries can change the patients represented and the patterns visible. Extreme values may also reflect meaningful observations.

We will examine how missing value handling affects sample composition, distributions and variable relationships in the public Chronic Kidney Disease (CKD) dataset. Transformations and scaling will provide additional visual comparisons. The project connects preprocessing decisions with descriptive statistics and visualization.

Research Questions

1. Which variables have the most missing values, and how does deleting incomplete records change the composition of the CKD and non-CKD groups?
2. How do complete-case analysis and simple imputation affect the distributions and associations of selected clinical measurements?
3. How do skewness, extreme observations and measurement scales influence the interpretation of individual-variable and multivariate plots?

Dataset Description

We will use the public Chronic Kidney Disease dataset from the UCI Machine Learning Repository [1,2]. Each row represents one patient record.

Dataset characteristic	Description
Size	400 patient records and 24 explanatory variables, plus one class label
Class composition	250 CKD records and 150 non-CKD records
Numerical measurements	Examples include age, blood pressure, blood glucose, blood urea, serum creatinine and hemoglobin
Categorical and coded variables	Examples include hypertension, diabetes, red blood cell assessment, and albumin and sugar grades
Data characteristics	Missing values, whitespace inconsistencies and different measurement scales

A preliminary audit of all 24 explanatory variables retained 158 complete records: 43 CKD and 115 non-CKD. The CKD share therefore changed from 62.5% in the full dataset to 27.2% after complete-case deletion [2]. These are sample proportions, not population disease prevalence. This observed change motivates our investigation of sample representation.

Its manageable size and mix of variable types support transparent inspection. The class label enables comparisons of how processing affects the two groups.

EDA and Data Visualization Plan

Data cleaning and missing value handling

We will preserve the source data and audit variable types, documented units, missing markers, category spellings, numerical conversions and duplicate rows. Whitespace will be normalized before missingness is assessed. Coded measurements will follow the data dictionary rather than automatically being treated as continuous variables. The diagnosis label will be excluded from explanatory-feature calculations.

We will compare three views: available observed values for each variable, the complete-case subset, and the full dataset with median imputation for numerical variables and mode imputation for categorical variables. Imputation will not use the diagnosis label. Indicators will identify originally missing entries, and each plot will report its sample size. Changes involving complete cases will be interpreted alongside the change in patient composition.

Summary statistics and extreme observations

The main analysis will focus on hemoglobin, serum creatinine, blood urea, blood glucose and blood pressure. We will report available sample sizes, means, medians, standard deviations and interquartile ranges, with group summaries where useful. Extreme values will be flagged and inspected; removal will require a documented data-quality reason. For suitable positively skewed measurements, original and logarithmic views will be compared, with transformations and units clearly stated.

Data visualization

We will use Python to prepare the data and create plots. Distribution comparisons will use consistent axes and histogram bins or empirical cumulative distribution curves. The planned visualizations connect directly to the research questions.

Visualization	Purpose
Missing-rate bars and a missingness matrix	Locate missing values and identify entries that are missing together
Group-count and proportion bars	Show sample loss and changes in CKD representation after deletion
Aligned histograms and cumulative distribution curves	Compare observed, complete-case and imputed distributions
Box plots and selected scatter plots	Examine group differences, extreme values and variable relationships
Correlation and patient-by-feature heatmaps	Summarize associations and compare measurements with different units

Correlation analysis and scale effects

Pearson and Spearman correlations will describe complementary aspects of relationships among selected numerical variables. We will report the sample size for each comparison and examine whether relationships persist across missing value strategies. Transformation and scaling comparisons will hold the patient set and other processing steps fixed wherever possible. Ordinary z-score standardization does not change Pearson correlations for fixed observations; its effects will be examined in patient-by-feature heatmaps and, optionally, PCA.

Other Methods (Optional)

Principal component analysis (PCA) may supplement the main plots. Using the same median-imputed records and a fixed set of numerical features, we will compare PCA before and after standardization

through score plots, explained variance and feature loadings. PCA will help illustrate scale effects, and visible groupings will be interpreted as exploratory patterns.

Expected Outcomes

We will produce a data-quality summary, a record of preprocessing decisions and matched visual comparisons. The analysis will show how complete-case deletion changes sample representation, whether simple imputation creates concentrations or changes distributional spread, and which selected relationships remain stable. We will evaluate processing choices through sample retention, visual interpretability and the consistency of descriptive findings. Conclusions will concern associations and processing effects within this dataset.

References

[1] Rubini, L., Soundarapandian, P., & Eswaran, P. (2015). Chronic Kidney Disease [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5G020

[2] UCI Machine Learning Repository. Chronic Kidney Disease official data CSV. https://archive.ics.uci.edu/static/public/336/data.csv (accessed 24 October 2026).